Dual-view Atomic and Fragment Graph Representation Learning for Drug Response Prediction

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Abstract. Drug response prediction (DRP) plays an important role in precision oncology by enabling the estimation of cancer cell sensitivity to therapeutic compounds. Recent graph neural network (GNN)-based approaches have achieved promising performance by representing drugs as molecular graphs; however, most existing methods primarily focus on atom-level structures and often overlook higher-order chemical semantics encoded in functional fragments. To address this limitation, we propose DAF-DRP, a dual-view framework that jointly learns atomic- and fragment-level molecular representations for drug response prediction. Specifically, each drug is modeled using two complementary graph views: an atomic graph capturing fine-grained local chemical connectivity and a fragment graph representing higher-order structural motifs derived from molecular decomposition. Separate Graph Isomorphism Networks (GINs) are employed to encode the two views, while genomic profiles of cancer cell lines are processed using a multilayer perceptron. The resulting representations are integrated through hierarchical fusion strategies, including concatenation, weighted addition, and cross-attention mechanisms. Experimental results on the CCLE and gCSI benchmark datasets demonstrate that DAF-DRP consistently outperforms established baselines such as GraphDRP, DeepCDR, HiDRA, and UNO across RMSE, PCC, and R^2 metrics. In particular, the combination of weighted addition and attention-based fusion achieves the best overall performance, highlighting the effectiveness of integrating fine-grained atomic information with fragment-level chemical priors for robust drug response prediction.

1 Introduction

Drug Response Prediction (DRP) is a cornerstone of computational personalized medicine, aiming to estimate the sensitivity of cancer cell lines to specific drugs [1]. In recent years, deep learning approaches, particularly Graph Neural Networks (GNNs), have become the dominant paradigm for DRP by modeling drugs as molecular graphs and cell lines via genomic features [2, 3].

Early deep learning methods primarily relied on Multi-Layer Perceptrons (MLPs) or Convolutional Neural Networks (CNNs) to process simplified molecular representations such as SMILES strings or fingerprint vectors [4]. While these methods established baseline performance, they often fail to capture the complex topological structure of molecules. To address

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this, graph-based approaches like GraphDRP [1] and DeepCDR[2] were introduced, utilizing GNNs to encode atomic interactions directly from molecular graphs. These models demonstrated superior performance by preserving local chemical connectivity. Subsequently, more advanced architectures like HiDRA [3] incorporated attention mechanisms to weigh the importance of different genes and substructures, while UNO [4] focused on cross-study generalization through unified normalization strategies.

We hypothesize that existing atom-level representations are suboptimal because they lack explicit chemical fragment priors. Since functional groups (fragments) are critical determinants of chemical activity [5], models operating exclusively at the atomic level may fail to fully capture the intrinsic relationship between high-order structural motifs and molecular properties. This motivates our approach to integrate fragment-level semantics alongside atomic details.

We propose DAF-DRP, a dual-view framework combining atomic and fragment molecular graphs with genomic profiles. By employing Graph Isomorphism Networks (GIN) to encode both local atomic connectivity and global fragment topology, DAF-DRP captures complementary chemical semantics. We integrate these representations with cell line data via hierarchical fusion operators. While fragment-aware molecular encoders have been explored in general molecular property prediction [5, 6], and hierarchical GNNs typically obtain coarse-grained representations by pooling nodes within a single graph, DAF-DRP differs in two respects specific to the DRP setting. First, the atomic and fragment views are encoded by two structurally and parametrically independent GIN encoders ($\theta_a \neq \theta_f$) operating on two distinct graphs G_a and G_f, rather than by pooling a shared graph hierarchically; this avoids conflating fine-grained atomic signal with coarse-grained fragment signal during message passing. Second, to our knowledge, dual-scale (atom+fragment) molecular graph representations have not been previously combined with cell-line genomic profiles for drug response prediction; prior DRP models [1, 2, 3, 4] operate exclusively at the atomic or descriptor level. DAF-DRP is thus best characterized as an adaptation and combination of dual-view molecular encoding with genomics-aware fusion, rather than a new graph operator. Experiments on CCLE [7] and gCSI [8] show that DAF-DRP outperforms state-of-the-art baselines, including GraphDRP, DeepCDR, and UNO.

The main contributions of this work are summarized as follows:

- We propose a dual-view graph representation strategy that combines atomic and fragment-level information to incorporate explicit chemical priors.
- We design a hierarchical fusion mechanism that adaptively integrates drug structural features with genomic profiles using Concatenation, Weighted Addition, and Cross-Attention operators.
- We achieve consistent performance improvements over established baselines on two within-dataset benchmarks (CCLE and gCSI), across all three evaluation metrics.

2 Proposed Methods

Formally, let $\mathcal{D}$ and $\mathcal{C}$ denote the feature spaces for drugs and cell lines, respectively. The objective is to learn a mapping function $f : \mathcal{D} \times \mathcal{C} \to \mathbb{R}$ that predicts a continuous response score $\hat{y}$ given a drug representation $d \in \mathcal{D}$ and a cell line representation $c \in \mathcal{C}$: $\hat{y} = f(d, c; \theta)$, where θ represents the learnable parameters. As illustrated in Figure 1, our framework employs a dual-view strategy grounded in [9]. We decompose the drug molecule into two complementary views: a fine-grained *AtomGraph* and a coarse-grained *FragGraph*. These are processed by dedicated encoders alongside the cell line's genomic profile. The resulting embeddings are integrated via hierarchical fusion operators to capture interactions across atomic, structural, and genetic scales.

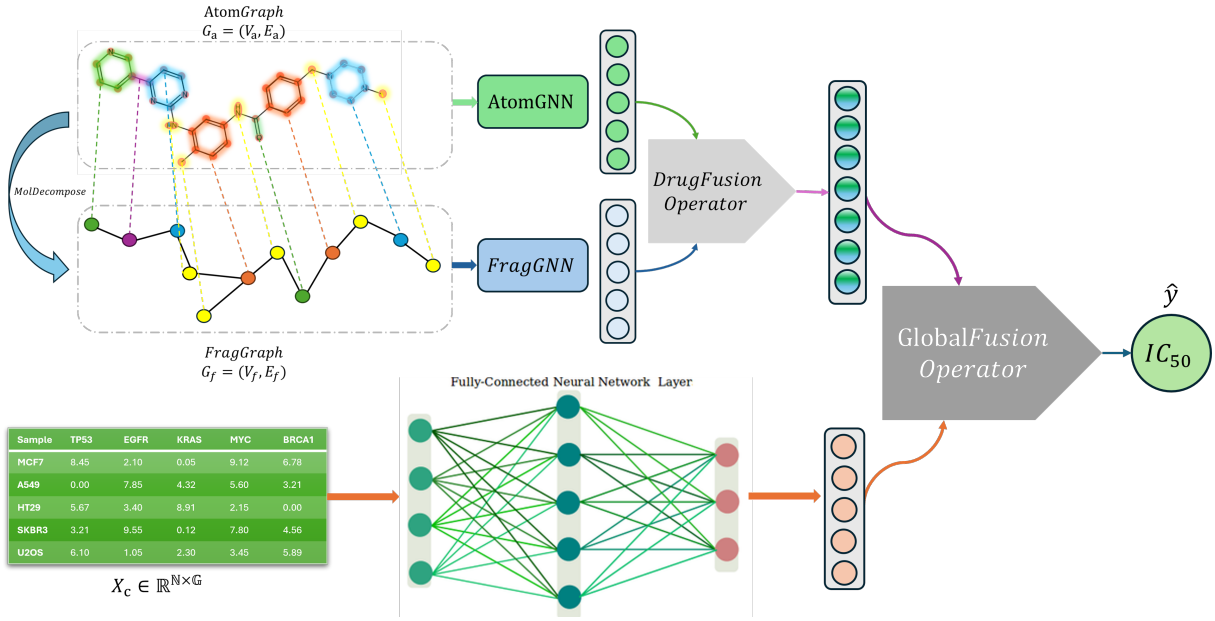


Figure 1: Overview of DAF-DRP architecture. The drug branch utilizes graph encoders (*AtomGNN*, *FragGNN*), while the cell line branch employs a genomic encoder. Representations are fused hierarchically to predict $\hat{y}$.

2.1 Drug Representation Learning via Dual-View Graphs

To comprehensively capture molecular properties, we represent each drug using two distinct graph structures derived from Principal Subgraph Mining [10, 9].

Molecular and Fragment Graph Decomposition. The fine-grained view is defined as the *AtomGraph* $G_a = (V_a, E_a)$, where nodes $v \in V_a$ represent atoms with features $\mathbf{x}_v^{(a)}$, and edges $(u, v) \in E_a$ represent chemical bonds with features $\mathbf{e}_{uv}$. This graph preserves precise local atomic connectivity.

To model higher-order structural dependencies, we construct the coarse-grained *FragGraph* $G_f = (V_f, E_f)$. Using the existing vocabulary of prevalent fragments $\mathcal{S} = \{S_1, \ldots, S_K\}$ from previous work [9], each molecule G_a is decomposed into non-overlapping fragments $\mathcal{F} = \{S^{(1)}, \ldots, S^{(M)}\}$ such that $\bigcup S^{(k)} = V_a$. The fragment graph $G_f = \text{MolDecompose}(G_a; \mathcal{S})$ is defined as:

- **Nodes (V_f):** Each node $u_k \in V_f$ corresponds to a fragment $S^{(k)} \in \mathcal{F}$. Node features are initialized via a learnable lookup table indexed by the fragment's identity in $\mathcal{S}$.
- **Edges (E_f):** An edge exists between fragments $S^{(i)}$ and $S^{(j)}$ if and only if they are connected in G_a:

$$E_f = \left\{(i, j) \mid \exists u \in S^{(i)}, v \in S^{(j)} \text{ s.t. } (u, v) \in E_a\right\}. \tag{1}$$

The fragment vocabulary $\mathcal{S}$ used in this work follows the extraction procedure of [9] and contains $|\mathcal{S}| = 798$ frequent substructures. By construction, $\mathcal{S}$ yields a non-overlapping partition of V_a , so no explicit conflict-resolution step is required for overlapping fragments. Atoms that do not match any entry in $\mathcal{S}$ are retained as singleton fragments, ensuring $\bigcup S^{(k)} = V_a$ holds for every molecule regardless of vocabulary coverage. Each fragment node is represented by a 128-dimensional embedding, initialized from a learnable lookup table of size 798×128 and indexed by the fragment's identity in $\mathcal{S}$. This abstraction reduces the molecule to a topological skeleton of functional groups, enabling the learning of global patterns invariant to local perturbations.

Graph Encoders. To encode the structural information from both views, we instantiate two separate Graph Neural Networks: *AtomGNN* for the atomic graph G_a and *FragGNN* for the fragment graph G_f. Both encoders adopt the Graph Isomorphism Network (GIN) [11] as their backbone architecture. GIN is selected due to its theoretical guarantee of maximal expressive power, being equivalent to the Weisfeiler-Lehman (WL) graph isomorphism test, which allows it to effectively distinguish diverse molecular topologies that less powerful GNNs might map identically.

Formally, let GIN_θ denote a GIN network with parameters θ. We define the encoding process for each view independently to prevent interference between local atomic features and global fragment patterns. The final latent representations are obtained as:

$$\mathbf{z}_a = \textit{AtomGNN}(G_a; \theta_a) = \text{SumPool}\left(\text{GIN}_{\theta_a}(G_a)\right), \tag{2}$$

$$\mathbf{z}_f = \textit{FragGNN}(G_f; \theta_f) = \text{SumPool}\left(\text{GIN}_{\theta_f}(G_f)\right) \tag{3}$$

where θ_a and θ_f are independent learnable parameters for the atomic and fragment branches, respectively. $\text{SumPool}(\cdot)$ represents the sum pooling that aggregates node-level embeddings into a fixed-size graph-level vector $\mathbf{z} \in \mathbb{R}^d$. This architecture design ensures that the model captures complementary chemical semantics: $\mathbf{z}_a$ preserves fine-grained local connectivity, while $\mathbf{z}_f$ encapsulates high-order functional group interactions.

2.2 Cell Line Genomic Feature Representation

Parallel to the drug branch, the cell line c is characterized by its gene expression profile. Let $\mathbf{x}_c \in \mathbb{R}^G$ denote the vector of log-transformed expression levels for G genes. To project this high-dimensional sparse data into a dense latent space suitable for fusion, we employ a Multi-Layer Perceptron (FC_{ge}) like in [1]:

$$\mathbf{h}_c = \text{FC}_{ge}(\mathbf{x}_c) = \sigma_L(\mathbf{W}_L \cdots \sigma_1(\mathbf{W}_1\mathbf{x}_c + \mathbf{b}_1) \cdots + \mathbf{b}_L), \tag{4}$$

where $\mathbf{W}_l, \mathbf{b}_l$ are learnable parameters and σ_l denotes non-linear activations. The resulting embedding $\mathbf{h}_c \in \mathbb{R}^{d_{cell}}$ distills critical genetic markers relevant to drug sensitivity.

2.3 Hierarchical Fusion Operators

We integrate the heterogeneous embeddings via two stages: *DrugFusion* ($\mathcal{O}_{drug}$) combining $\mathbf{z}_a$ and $\mathbf{z}_f$, and *GlobalFusion* ($\mathcal{O}_{global}$) merging the result with $\mathbf{h}_c$. For any pair of inputs $\mathbf{a}, \mathbf{b}$, we evaluate three fusion mechanisms:

1. **Concatenation:** Preserves full information via stacking and projection: $\mathcal{O}^{concat}(\mathbf{a}, \mathbf{b}) = \mathbf{W}_{proj}[\mathbf{a} \parallel \mathbf{b}] + \mathbf{b}_{proj}$.

2. **Weighted Element-wise Addition:** Enforces latent space alignment while adaptively balancing the contribution of each view via a globally shared learnable scalar $\alpha \in [0, 1]$:

$$\mathcal{O}^{\alpha}(\mathbf{a}, \mathbf{b}) = \alpha\mathbf{a} + (1 - \alpha)\mathbf{b}. \tag{5}$$

3. **Cross-Attention:** Dynamically weights contributions by treating $\mathbf{a}$ as Query and $\mathbf{b}$ as Key/Value:

$$\mathcal{O}^{attn}(\mathbf{a}, \mathbf{b}) = \text{Softmax}\left(\frac{\mathbf{a}\mathbf{W}_Q(\mathbf{b}\mathbf{W}_K)^T}{\sqrt{d_k}}\right)(\mathbf{b}\mathbf{W}_V). \tag{6}$$

We use a single attention head with $d_k = d = 128$, and $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d \times d}$ are learnable projection matrices specific to each fusion stage.

Finally, the joint representation $\mathbf{h}_{final} = \mathcal{O}_{global}(\mathcal{O}_{drug}(\mathbf{z}_a, \mathbf{z}_f), \mathbf{h}_c)$ is passed through a prediction head to output the response score:

$$\hat{y} = \text{MLP}_{pred}(\mathbf{h}_{final}), \tag{7}$$

optimized end-to-end via Mean Squared Error (MSE) loss.

3 Experiments and Results

3.1 Experimental Setup

Implementation and training. DAF-DRP is implemented based on the IMPROVE benchmarking framework [12] to ensure identical data splits, preprocessing, and evaluation protocols across all compared models. Table 1 summarizes the architectural and optimization hyperparameters, which are held fixed across all six fusion configurations (CC, WC, CA, WA, AC, AA) to isolate the effect of the fusion strategy. All experiments use a fixed random seed (42) and are run on a single GPU.

Table 1: Architectural and training hyperparameters.

Component	**Value**
AtomGNN / *FragGNN* layers (L_a, L_f)	5 / 5
GNN hidden dimension	128
Fragment vocabulary size	798
Embedding dimension (d)	128
Cell-line MLP hidden dimension (d_{cell})	128
Dropout rate	0.2
Prediction head	$128 \to 64 \to 1$
Optimizer	Adam
Learning rate	1×10^{-4}
Loss / early-stopping metric	MSE
Early-stopping patience	20
Batch size	256
Max. epochs	150
Random seed	42

Data. Following [12], response labels y, cancer gene expression profiles $\mathbf{x}_c$, and drug SMILES are drawn from the standard benchmark files. Gene expression is restricted to the LINCS L1000 landmark set (958 genes) and standardized (z-score) prior to training. Train/validation/test partitions follow the predefined 10-fold split files, ensuring all models are evaluated on identical splits.

3.2 Evaluation Criteria

Following [12], we evaluate the performance using three standard metrics: Root Mean Squared Error (RMSE), Pearson Correlation Coefficient (PCC), and the Coefficient of Determination (R^2). RMSE quantifies the prediction error magnitude in the same unit as the response variable (where lower values $\downarrow$ are better). PCC measures the linear correlation between predicted and observed responses, while R^2 indicates the proportion of variance explained by the model. Higher values ($\uparrow$) for PCC and R^2 indicate superior predictive capability.

3.3 Performance Comparison

We conducted performance comparison analyses using 10-fold cross-validation. We assess six fusion configurations of our proposed DAF-DRP on CCLE [7] (Table 2) and gCSI [8] (Table 3). These datasets were selected following [12], as their molecular structures are compatible with vocabulary fragment extraction [9], ensuring consistent fragment vocabulary across experiments. We examine different combinations of fusion mechanisms on the DrugFusion and GlobalFusion operators, abbreviated as: Concatenation (C), Weighted Addition (W), and Cross-Attention (A). We compares these configurations with baselines GraphDRP [1], DeepCDR [2], HiDRA [3], and UNO [4] which are re-trained with their default configurations from the same benchmarking suite, under identical splits and preprocessing Overall, DAF-DRP consistently outperforms established baselines across both datasets. Notably, the configuration combining Weighted Addition for drug-level fusion and Cross-Attention for global integration achieves the highest predictive accuracy, demonstrating the effectiveness of adaptively balancing atomic-fragment semantics with genomic context.

Table 2: **Performance comparison on CCLE.**

	Method	RMSE ↓	PCC ↑	R^2 ↑
Baseline	GraphDRP [1]	0.618 ± 0.010	0.865 ± 0.008	0.746 ± 0.011
	DeepCDR [2]	0.592 ± 0.008	0.878 ± 0.007	0.766 ± 0.009
	HiDRA [3]	0.604 ± 0.009	0.871 ± 0.008	0.756 ± 0.010
	UNO [4]	0.558 ± 0.007	0.892 ± 0.006	0.796 ± 0.008
Ours	DAF-DRP$_{CC}$	0.573 ± 0.009	0.885 ± 0.007	0.785 ± 0.009
	DAF-DRP$_{WC}$	0.561 ± 0.008	0.894 ± 0.006	0.798 ± 0.008
	DAF-DRP$_{CA}$	0.555 ± 0.007	0.898 ± 0.005	0.802 ± 0.007
	DAF-DRP$_{WA}$	$\mathbf{0.543 \pm 0.006}$	$\mathbf{0.905 \pm 0.005}$	$\mathbf{0.815 \pm 0.006}$
	DAF-DRP$_{AC}$	0.549 ± 0.007	0.901 ± 0.005	0.808 ± 0.007
	DAF-DRP$_{AA}$	0.546 ± 0.006	0.903 ± 0.005	0.811 ± 0.006

Table 3: **Performance comparison on gCSI.**

	Method	RMSE ↓	PCC ↑	R^2 ↑
Baseline	GraphDRP [1]	0.644 ± 0.011	0.858 ± 0.009	0.736 ± 0.012
	DeepCDR [2]	0.654 ± 0.010	0.848 ± 0.010	0.720 ± 0.013
	HiDRA [3]	0.667 ± 0.012	0.841 ± 0.010	0.711 ± 0.013
	UNO [4]	0.604 ± 0.009	0.880 ± 0.007	0.774 ± 0.010
Ours	DAF-DRP$_{CC}$	0.648 ± 0.010	0.855 ± 0.009	0.731 ± 0.012
	DAF-DRP$_{WC}$	0.610 ± 0.009	0.882 ± 0.007	0.778 ± 0.010
	DAF-DRP$_{CA}$	0.628 ± 0.010	0.870 ± 0.008	0.753 ± 0.011
	DAF-DRP$_{WA}$	$\mathbf{0.598 \pm 0.008}$	$\mathbf{0.892 \pm 0.006}$	$\mathbf{0.792 \pm 0.009}$
	DAF-DRP$_{AC}$	0.607 ± 0.009	0.885 ± 0.007	0.786 ± 0.010
	DAF-DRP$_{AA}$	0.602 ± 0.008	0.888 ± 0.006	0.789 ± 0.009

Conclusion

We proposed DAF-DRP, a dual-view framework integrating atomic and fragment molecular graphs with genomic profiles to capture complementary chemical semantics. Unlike atom-level

GNNs, our approach explicitly incorporates fragment priors to model higher-order structural motifs. Experiments on CCLE and gCSI demonstrate that DAF-DRP consistently outperforms baselines like GraphDRP, DeepCDR, and UNO. The Weighted Addition and Cross-Attention configuration yielded the best results, confirming that balancing atomic precision with fragment abstraction enhances prediction accuracy.

Despite the promising results, several limitations remain. First, the current framework utilizes predefined fragment vocabularies, which may not fully capture all biologically relevant molecular substructures. Second, the model currently relies only on gene expression profiles and does not integrate other omics modalities such as mutation, methylation, or proteomics data. In addition, the experiments were limited to within-dataset evaluation, and further cross-dataset generalization studies are needed to assess robustness in more heterogeneous settings.

Future work will directly target these gaps: an atom/fragment ablation study, paired significance testing across folds, comparison-matched hyperparameter tuning against more recent baselines, cross-dataset generalization experiments, adaptive fragment discovery, and multimodal omics integration. These additions, planned for an extended version of this work, will allow a more rigorous assessment of the value of fragment-level priors for drug response prediction.

Conflict of interests

The authors declare no competing interests.

Availability of data and software code (optional and strongly suggested)

Our software code is publicly available at the following GitHub repository: `https://github.com/khiemducdoan/DAF-DRP.git`.

References

[1] Tuan Nguyen, Giang TT Nguyen, Thin Nguyen, and Duc-Hau Le. Graph convolutional networks for drug response prediction. *IEEE/ACM transactions on computational biology and bioinformatics*, 19(1):146–154, 2021.

[2] Qiao Liu, Zhiqiang Hu, Rui Jiang, and Mu Zhou. Deepcdr: a hybrid graph convolutional network for predicting cancer drug response. *Bioinformatics*, 36(Supplement_2):i911–i918, 2020.

[3] Iljung Jin and Hojung Nam. Hidra: hierarchical network for drug response prediction with attention. *Journal of Chemical Information and Modeling*, 61(8):3858–3867, 2021.

[4] Fangfang Xia, Jonathan Allen, Prasanna Balaprakash, Thomas Brettin, Cristina Garcia-Cardona, Austin Clyde, Judith Cohn, James Doroshow, Xiaotian Duan, Veronika Dubinkina, et al. A cross-study analysis of drug response prediction in cancer cell lines. *Briefings in bioinformatics*, 23(1):bbab356, 2022.

[5] Jiaxi Wang, Yaosen Min, Miao Li, and Ji Wu. Fragformer: A fragment-based representation learning framework for molecular property prediction. *Transactions on Machine Learning Research*, 2025.

[6] Yinghui Jiang, Shuting Jin, Xurui Jin, Xianglu Xiao, Wenfan Wu, Xiangrong Liu, Qiang Zhang, Xiangxiang Zeng, Guang Yang, and Zhangming Niu. Pharmacophoric-constrained heterogeneous graph transformer model for molecular property prediction. *Communications Chemistry*, 6(1):60, 2023.

[7] Jordi Barretina, Giordano Caponigro, Nicolas Stransky, Kavitha Venkatesan, Adam A Margolin, Sungjoon Kim, Christopher J Wilson, Joseph Lehár, Gregory V Kryukov, Dmitriy Sonkin, et al. The cancer cell line encyclopedia enables predictive modelling of anticancer drug sensitivity. *Nature*, 483(7391):603–607, 2012.

[8] Christiaan Klijn, Steffen Durinck, Eric W Stawiski, Peter M Haverty, Zhaoshi Jiang, Hanbin Liu, Jeremiah Degenhardt, Oleg Mayba, Florian Gnad, Jinfeng Liu, et al. A comprehensive transcriptional portrait of human cancer cell lines. *Nature biotechnology*, 33(3):306–312, 2015.

[9] Kha-Dinh Luong and Ambuj Singh. Fragment-based pretraining and finetuning on molecular graphs. In *Advances in Neural Information Processing Systems*, 2023.

[10] Xiangzhe Kong, Wenbing Huang, Zhixing Tan, and Yang Liu. Molecule generation by principal subgraph mining and assembling. *Advances in Neural Information Processing Systems*, 35:2550–2563, 2022.

[11] Keyulu Xu, Weihua Hu, Jure Leskovec, and Stefanie Jegelka. How powerful are graph neural networks? *arXiv preprint arXiv:1810.00826*, 2018.

[12] Alexander Partin, Priyanka Vasanthakumari, Oleksandr Narykov, Andreas Wilke, Natasha Koussa, Sara E Jones, Yitan Zhu, Jamie C Overbeek, Rajeev Jain, Gayara Demini Fernando, et al. Benchmarking community drug response prediction models: datasets, models, tools, and metrics for cross-dataset generalization analysis. *Briefings in Bioinformatics*, 27(1):bbaf667, 2026.